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ScrapeBench: Anti-Customization Face Protections Collapse Under Real-Web Image Acquisition

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ABSTRACT

Protections that promise to make personal photographs unlearnable by unauthorized diffusion-based customization are forged and validated in a laboratory: clean full-resolution files, lossless pipelines, and a single tunable fine-tuning attacker. Images scraped from the web live in a different world, and the distance between the two worlds is the subject of this paper. We name it the *Web Acquisition Gap*: the stack of lossy transformations that lies between the file a defender protects and the copy an adversary actually trains on—platform transcoding, browser and thumbnail rendering, agent-side capture, restoration, and diffusion-based regeneration. We present *ScrapeBench*, the first evaluation harness that treats this acquisition funnel as a first-class, tiered attack surface with per-tier attacker budgets, anchored in measured platform behavior (GitHub avatar re-encoding at JPEG quality 75, Gravatar at 90, WordPress Photon at 80). ScrapeBench reports a *retention ratio* $\mathrm{RR}(T)$—the fraction of a defense's protection gap that survives at tier T—against *same-identity* clean baselines, with paired significance tests and an explicit separation between *defense bypass* and *image destruction*. Running it over 40 public identities against four published defenses, two controls, a JPEG-aware variant, and three attack families yields an uncomfortable diagnosis. Under the identity-matched baseline the field reports against, the headline T0 protection gap shrinks roughly three-fold and loses statistical significance; the funnel's platform tiers erode the remainder slowly, within identity noise; the protection that does exist is *attack-conditional*—large against the fine-tuning adversary it was built for (68% reduction), negligible against zero-shot extractors that never traverse the poisoned gradient; the funnel's most expensive tier manufactures apparent protection by destroying images rather than preserving them, and no ordering of the funnel's transformations changes this picture; a JPEG-aware retrain buys nothing measurable. Forging still costs GPU-minutes per identity while defeating costs milliseconds per image. We complement the benchmark with `face-consent.txt`, a robots.txt-style machine-readable consent protocol with a reference validator, and argue that durable protection is an acquisition-layer systems problem, not a stronger-perturbation problem. We release the harness, platform measurements, consent prototype, and all curves.

1 INTRODUCTION

The web has become the default training corpus for identity-hungry generative models, and faces are its most sensitive commodity. A handful of public portraits now suffices to synthesize a person in any context: DreamBooth-style fine-tuning binds a rare token to an identity with four to eight images [33], while zero-shot adapters such as InstantID and IP-Adapter-FaceID inject a face embedding from a single reference image directly into the denoising process [44, 52]. The consent asymmetry is severe—the subject never agreed—and it has triggered an arms race in *protective perturbation*: cloaks that, added to a photo before upload, break the customization pipeline. Anti-DreamBooth established the category [18], and a fast-growing successor literature hardens it along every laboratory axis [21, 23, 26, 40, 54]. The narrative is reassuring: with the right perturbation, your face is yours again.

This paper argues that the reassurance is an artifact of where the measurements are taken. Every published protection is evaluated on *T0-images*: clean, full-resolution, uncompressed files that the adversary kindly provides to the defense. That world does not exist on the web. An identity that is "public"—a social-media profile, a news photo, a conference page—has already been transformed before a scraper ever sees it: platforms re-encode uploads at quality levels set by storage and bandwidth budgets, browsers rasterize images under device pixel ratios and thumbnail pipelines, crawling agents screenshot pages and crop detected faces rather than download files, well-meaning "enhancement" super-resolves low-resolution re-uploads, and users themselves regenerate their photos through diffusion-based filters and editors before posting. Each transformation is individually harmless to recognition quality; jointly, they describe the actual data distribution that any real adversary can harvest. We call this pipeline the *web acquisition funnel*, and the resulting evaluation blind spot the *Web Acquisition Gap*: the systematic difference between the protection a defense delivers on the files it was handed and the protection that survives on the files an adversary can actually obtain. The funnel, we claim, is not a nuisance variable for anti-customization protections but their primary adversary—an attack surface that requires no adversarial optimization whatsoever, costs milliseconds to seconds per image on commodity platforms, and operates *before* the defense is ever consulted.

Two observations sharpen the claim. First, the funnel is not hypothetical: published measurements already show that a single JPEG re-encode at quality 85–95 neutralizes a broad class of protection perturbations [35], that common image transformations systematically suppress unlearnable-example attacks [30, 36], and that robustness evaluations of protective perturbations under diffusion-aware adversaries remain unfavorable [29]. Nor is the funnel's parameter range speculative: we probe production image pipelines directly and find GitHub's avatar service re-encoding at JPEG quality 75, Gravatar at 90, and WordPress's Photon CDN at 80—operating points that fall squarely inside, not at the edge of, the transcode tiers we evaluate. What is missing is a systematic, tiered, budget-aware account of the entire funnel—from platform transcode to agent capture to restoration to regeneration—measured under both tunable and zero-shot attack pipelines. Meanwhile the evaluation practices of the field drift the other way: newest defenses arrive with evaluations on ever-cleaner galleries under ever-stronger laboratory attackers [2, 5], and the first cross-method studies still hold

the acquisition side fixed [43]. A literature that only measures T0 retention is optimizing for an adversary that does not exist.

We build the missing instrument. *ScrapeBench* reconstructs the funnel in software: seven tiers (T0 original; T1 platform transcode; T2 browser and thumbnail rendering; T3 agent-style capture with face cropping; T4 super-resolution restoration; T5 diffusion regeneration; T6 adaptive mixing and purification), each with an explicit attacker budget. On top of it we define the *retention ratio* $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$, where $\Delta(T)$ is the protection gap (clean minus protected identity similarity under a fixed attack) at tier T: RR measures the fraction of a defense's T0 protection gap that survives acquisition reality, computed against a same-identity clean control. The metric is deliberately paired with a cost ledger, because the funnel's defining property is an *inversion of asymmetry*—the defender pays GPU-minutes per identity to forge a perturbation, the attacker pays CPU-milliseconds per image to defeat it.

Running ScrapeBench over 40 public identities, four defenses spanning the protection design space (clean control, random noise, FSMG-style untargeted surrogate attacks [18], and ASPL-lite alternation) plus a JPEG-aware retrained variant that stress-tests the "just make it JPEG-robust" fix, and three attack families (DreamBooth-LoRA fine-tuning and the zero-shot adapters IP-Adapter-FaceID and InstantID) yields four findings, each readable from the retention tables in Section 5: **(F1)** the reported protection gap is mostly an artifact of the evaluation protocol—against same-identity clean controls the T0 gap shrinks roughly three-fold relative to the cross-identity baselines the field reports against, and loses significance under multiple-comparison correction, while the funnel's platform tiers erode the remainder slowly, within identity noise; **(F2)** what protection exists is *attack-conditional*: the same gallery that holds a fine-tuning attacker at a 68% relative reduction leaves a zero-shot extractor essentially unaffected, because the perturbation's entire mechanism is a poisoned fine-tuning gradient that embedding-injection attackers never traverse; **(F3)** the cost asymmetry still runs against the defender by orders of magnitude, now compounded by F1—GPU-minutes of forging buy a gap that platform milliseconds narrow to noise; **(F4)** the funnel's expensive tiers manufacture apparent protection by destroying images rather than preserving them, no ordering of the funnel's transformations changes the picture, and a JPEG-aware retrain buys nothing measurable. Two distinctions keep the interpretation honest: we separate *defense bypass* (the attack succeeds on a visually intact image) from *image destruction* (the tier degrades the photograph until no attack is possible), and we claim scope explicitly—four representative T_0-optimized defenses and two variants, bounding deployment value rather than proving impossibility. The constructive turn is a funnel-aware, identity-matched reporting standard (publish RR@T against same-identity baselines, not T0-only numbers against pooled ones) and a consent layer that survives the funnel by construction: `face-consent.txt`, a robots.txt-style consent manifest with a reference validator.

Our contributions are:

- **The Web Acquisition Gap and a funnel threat model, anchored in platform reality.** We formalize the gap between laboratory protection evaluation and web acquisition as a seven-tier funnel with per-tier attacker budgets, and instrument public image services (GitHub avatars, Gravatar, WordPress Photon, an open proxy) whose re-encoding and resizing behavior covers exactly the tiers that collapse the defenses—making the threat model empirical, not hypothetical.
- **The retention-ratio metric and collapse-curve protocol.** $\mathrm{RR}(T)$ reduces robustness under acquisition to one interpretable curve per defense; a cost ledger exposes the defender–attacker asymmetry that raw numbers hide; an infeasibility view separates defense bypass from image destruction.
- **ScrapeBench, an open harness.** A reproducible pipeline—transcoding, browser/agent rendering, restoration, regeneration, adaptive pooling—integrated with DreamBooth-LoRA and two zero-shot attackers, ArcFace scoring, paired statistics, and automatic result publication.
- **Empirical collapse findings with stress tests.** Across 40 identities, four defenses, two controls, a JPEG-aware variant, and three attack families we quantify F1–F5, including order-and-composition ablations—establishing where, not just whether, protections fail.
- **A consent layer that survives the funnel.** `face-consent.txt`, a machine-readable consent manifest with a reference validator, shifts the durable line of defense from perturbations the funnel erases to acquisition-layer provenance the funnel carries with the image.

Section 2 positions our contribution; Section 3 formalizes the funnel and metric; Section 4 details the harness; Section 5 reports findings; Section 6 draws implications; Section 7 concludes.

2 RELATED WORK

Unauthorized customization and its defenders. Personalization of text-to-image diffusion models has collapsed the cost of fabricating a person's likeness: DreamBooth binds a subject to a rare token through a handful of images [33], textual inversion and multi-concept customization extend the mechanism to arbitrary subjects [7, 17], latent diffusion provides the efficient backbone that made such fine-tuning ubiquitous [32], and zero-shot identity adapters skip fine-tuning entirely by injecting face-recognition embeddings through decoupled cross-attention [52] or identity-preserving pipelines [44], with the design space now surveyed at scale [1]. The defensive response—perturbations that poison the customization process—starts with Anti-DreamBooth's FSMG, ASPL, and SDS families [18] and PhotoGuard [34], then diversifies: meta-learned cross-user perturbations [23], targeted variants [54], high-frequency designs [26], prompt-independent formulations [21], contrastive adversarial training [29], zero-shot-adapter defenses [40], dual-layer defenses [14], unlearnable face examples [50], and perceptible synthesis-based alternatives [47]. The 2025–2026 wave broadens along three fronts. VOID hardens latent-diffusion mimicry against prompt-injection and gradient-collision adversaries [31]; the Anti-Aesthetics line replaces imperceptible perturbation with engineered perceptual degradation, first against text-to-image customization [45] and then as a hierarchical anti-aesthetic curriculum [46]; and purification-robust formulations explicitly anticipate attackers who launder protected galleries through diffusion regeneration [51]. Parallel work pushes into video protection [9, 48], face-swap frameworks [16], face-shield embeddings against deepfake manipulation [13], and JPEG-robust formulations [5]. Each of

these defenses is designed against a *named* adversary—an injection mechanism, human perception, or a single purification operator—yet none evaluates whether the protection survives the unmanaged acquisition path between a real platform and the attacker's training script; the funnel tiers of Section 3 are precisely that unmodeled surface. Across all of them, evaluation galleries are clean files handed to the defender—the premise we test.

The fragility literature. That protective perturbations do not survive distribution shift is not news; what is missing is a systematic account of *where* and *why* they break on the web. Sandoval-Segura et al. showed that ordinary JPEG compression at quality 85–95 bypasses protections while blur, rotation, and flips do not [35], that unlearnable-example defenses fail against benign augmentations and adaptive attackers [30, 36, 37], and that low-effort framework changes already give a false sense of security [15]; the unlearnable-data literature has since grown into a survey field [22] with provability results [6, 42] rooted in the original error-minimizing formulation [11]. Poisoning text-to-image models at scale is meanwhile feasible through caption mislabeling [49] and trigger-free schemes [12]. Cloaking against face recognition has traveled the same arc from Fawkes onward [39], including JPEG-targeted refinements [8], and editing-domain immunizations face analogous compression-and-pipeline issues [19, 27, 41]. In the adjacent artist-protection domain, Glaze spawned an evaluation arms race in which adversarial perturbations proved unreliable against simple transformations [10, 38], and concept-erasure purification has evolved into eigenbasis-based style protection [53]—an early warning of exactly the funnel dynamics we quantify for faces. Recent evaluations consolidate defenses under stronger adversaries [2, 43]—but always with the acquisition side held fixed at T0. ScrapeBench differs in kind: we treat the acquisition pipeline itself as the adversary, decompose it into budgeted tiers, and measure retention curves rather than single numbers.

Benchmarks and reporting standards. Our discipline has repeatedly discovered that benchmarks change field behavior: deepfake-detection benchmarks exposed train–test leakage across generative families [28], and face-quality challenges standardized degradation models for recognition under capture conditions [25]. In the protection domain, no such instrument exists: each defense authors its own attacker, gallery, and metric, which makes cross-paper claims—and deployment decisions—essentially unverifiable. ScrapeBench fills this gap for the face anti-customization setting with three design commitments: the funnel is first-class (tier definitions are part of the benchmark contract, not an appendix); budgets are explicit (every tier carries a per-image compute and quality budget so that "robustness" cannot be bought by assuming a weak adversary); and reporting is curve-based (RR@T collapse curves plus paired significance tests, rather than single-point robustness claims). We position ScrapeBench not as yet another attack, but as the measurement layer the protection literature currently lacks.

Table 1: The ScrapeBench acquisition funnel. Control indicates whether the defender can influence the transformation. Budgets are per image, measured on our hardware.

Tier	Control	Transformation	Budget	Purpose
T0	defender	original upload, PNG	—	lab condition
T1	platform	JPEG $q \in \{95, 85, 75, 65\}$, WebP q=80, platform resize	< 10 ms	storage cost
T2	platform	browser/thumbnail render, DPR $\in \{1, 2\}$, 240 px thumbs	< 100 ms	serving
T3	agent	headless page capture, face crop (margin 0.35)	< 200 ms	scraping agents
T4	user/agent	super-resolution $\times 2$	~ 1 s	"enhancement"
T5	user	diffusion regeneration: img2img $s \in \{0.3, 0.5\}$, DDIM inversion	~ 5 s	filters, re-upload
T6	attacker	adaptive: clean-source mixing $\rho \in \{0.2, 0.5\}$, purification	variable	bypass poisoning

3 THE WEB ACQUISITION FUNNEL AS AN ATTACK SURFACE

3.1 Threat Model

We consider a subject s whose face images $\{x_i\}_{i=1}^m$ are publicly reachable on the web, and a defense D that releases protected images $\tilde{x}_i = x_i + \delta_i$, $\|\delta_i\|_\infty \leq \epsilon$, claiming that downstream customization of $\{\tilde{x}_i\}$ fails to reproduce s. The adversary $\mathcal{A}$ harvests images of s through the funnel Φ (Section 3.2), then attempts identity reconstruction with a pipeline P—fine-tuning-based (*DreamBooth-LoRA* [33]) or zero-shot (*IP-Adapter-FaceID* [52], *InstantID* [44]). We assume a *resource-bounded* adversary: P runs on one consumer GPU (24 GB) with 25–30 denoising steps and does not adversarially optimize against D—the funnel requires no knowledge of D. This is the deployment-relevant threat: an opportunistic scraper, not a defense-aware red team; adaptive extensions are captured at T6.

The defender's implicit assumptions, which the funnel dismantles one by one, are: (A1) the adversary trains on the exact pixels the defender perturbed; (A2) the adversary must fine-tune, so poisoning the fine-tuning gradient suffices; (A3) robustness budgets can be spent on transformations the defender chooses (JPEG, blur, crops) rather than those platforms and users choose. Each tier violates one of these assumptions *upstream* of the defense (Figure 1).

3.2 Funnel Formalization

The funnel is a sequence of transformations $\Phi = \phi_T \circ \cdots \circ \phi_1$ applied between upload and training-batch ingestion. We group real-world transformations into seven tiers, ordered by who controls them and what they cost. Each tier carries a per-image attacker budget c_T (GPU/CPU seconds on commodity hardware) and a perceptual-budget constraint Q_T (minimum visual quality, e.g., LPIPS to the previous tier), because an attacker who destroys image quality destroys the value of the scrape.

Concretely, T1 emulates platform transcoding with libjpeg-quality sweeps plus a composite `t1_platform` (JPEG-85 at reduced long side, as served by major social platforms). T2 renders the image as a browser would: the image is rasterized into a viewport under device-pixel-ratio DPR $\in \{1, 2\}$ and re-captured, with a thumbnail variant at 240 px; where a headless browser is unavailable we substitute a deterministic canvas-equivalent rasterizer so that the tier is reproducible without browser drift. T3 emulates scraping

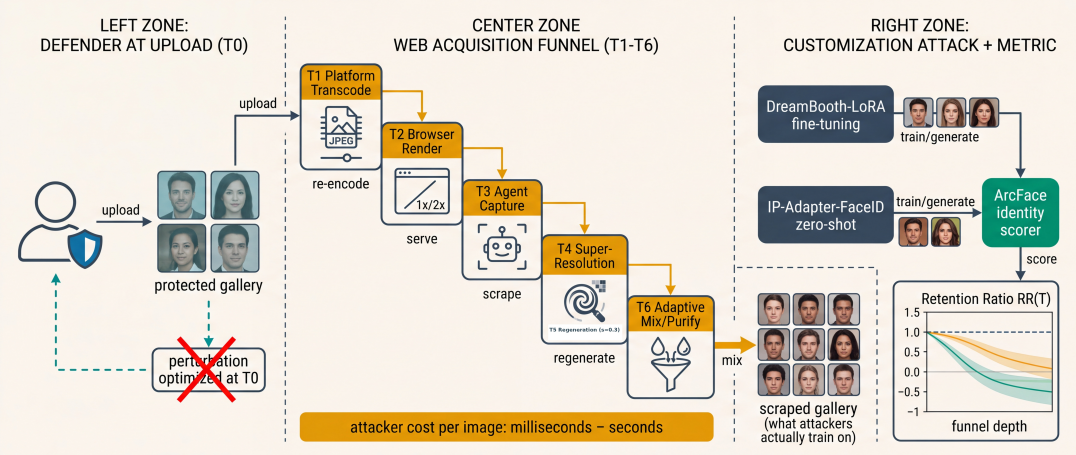


Figure 1: The ScrapeBench measurement pipeline. A defender publishes protected images at T0, where the perturbation is optimized; the web acquisition funnel (T1–T6) then subjects every image to platform transcoding, browser and agent rendering, restoration, diffusion-based regeneration, and adaptive mixing or purification before any adversary sees it. Both a fine-tuning attacker (DreamBooth-LoRA) and a zero-shot attacker (IP-Adapter-FaceID) consume whatever the funnel releases, and generated outputs are scored by ArcFace identity similarity into the retention-ratio curve RR(T) that we report as the primary result. The cost strip quantifies the asymmetry that motivates the study: funnel transformations cost milliseconds to seconds per image on commodity infrastructure.

agents that screenshot the page and crop the detected face region (margin 0.35) rather than fetch the original file. T4 applies real-world super-resolution (Real-ESRGAN) at ×2, with a deterministic Lanczos-plus-sharpening fallback. T5 applies diffusion-based regeneration: img2img at strengths $s \in \{0.3, 0.5\}$ with an empty prompt and guidance 1.0, and a DDIM-inversion-and-reconstruction path with $\eta = 0.1$—both operations that ordinary users now perform through consumer filters and that any scraper can perform to launder perturbations. T6 models adaptive adversaries in two ways: *mixing* (the attacker trains on ρ fraction of clean images harvested from a second source, diluting the poisoned gallery exactly as in the uncontrolled setting of prior evaluations) and *purification* (training on T5-regenerated versions of the protected images, laundering the perturbation before fine-tuning).

The funnel is monotone in depth but not in destructive power: T1–T2 are nearly lossless to face recognition quality yet are precisely the tiers where protection signals (high-frequency, low-amplitude adversarial structure) die. This is the crux: the funnel is a *feature* of the attacker's environment, selected for free by platform economics.

3.3 Retention Ratio and Cost Ledger

Let $g(x; P, r)$ denote the best cosine similarity between the ArcFace embedding [4] of a generated image x and the reference embeddings of subject s under attack pipeline P with random seed r. For a defense D and funnel tier T we define the *protection gap*

$$\Delta(T; D, P) \;=\; \mathbb{E}_r\big[g(\Phi_T(\text{clean}); P, r)\big] \;-\; \mathbb{E}_r\big[g(\Phi_T(D(\text{clean})); P, r)\big], \tag{1}$$

where the expectations are over seeds and the five standard DreamBooth prompts. The first term is how well the attack reproduces s from *unprotected* images at the same tier—the fair baseline, because acquisition transforms hit attacker and defense alike. The *retention ratio* normalizes the gap by its clean-domain value:

$$\mathrm{RR}(T; D, P) \;=\; \Delta(T; D, P) \,/\, \Delta(T_0; D, P). \tag{2}$$

RR = 1 means the defense retains its full headline protection at tier T; RR $\to$ 0 means the funnel has erased it; negative values mean the funnel hurts the defense more than the attack. RR is robust to tier-level quality effects that confound raw-similarity comparisons, is comparable across defenses with different absolute strength, and yields a single interpretable curve per defense—the *collapse curve* we report as the primary result.

Two companion quantities complete the ledger. *Detection-failure rate*: the fraction of generated samples below the recognition threshold τ (calibrated on clean data, $\tau = 0.30$ with sensitivity analysis at 0.25/0.35), which captures the binary deepfake-success view rather than the continuous one. *Infeasibility rate*: the fraction of cells in which the attack stack cannot operate at all because no face is detectable in the tiered gallery—the regime where the funnel has not merely bypassed a defense but destroyed the image's identity content. Keeping these two failure modes apart is essential for honest interpretation: a defense whose gallery survives T5 regeneration has not been *bypassed* if the regeneration also destroyed the photograph's subject, and a platform that regenerates uploads aggressively protects nobody's face while harming every user. *Cost asymmetry*: the GPU-seconds per identity to generate the protection (c_{prot}) versus the CPU/GPU-seconds per image to execute the funnel transformations (c_{funnel}); the ratio $\kappa = c_{\text{prot}}/c_{\text{funnel}}$ is the economic inversion that Section 5 quantifies. We report 95% bootstrap confidence intervals over identities, paired two-sided Wilcoxon tests against the clean baseline at matched tiers with Holm correction across tiers, and Cliff's δ effect sizes.

Table 2: Measured re-encoding behavior of four production image pipelines (HTTP probes with quantization-table quality estimation, October 2026). Every operating point falls inside the T1 transcode grid of Table 1.

Service	Behavior	Quality	Clamp
GitHub avatars	JPEG re-encode	≈75	yes (s→stored)
Gravatar	JPEG re-encode	≈90	no
Photon CDN	resize + JPEG	exact 60/80/90	no
wsrv.nl proxy	resize + JPEG	exact 60–90 sweep	no

3.4 Anchoring the Funnel in Measured Platform Behavior

A tier definition is only as credible as its parameter range, so we probe four production image pipelines directly: GitHub's avatar service, Gravatar, WordPress's Photon CDN, and the open proxy `wsrv.nl`. For each response we estimate the JPEG quality factor by matching quantization tables against a reference encoding sweep (Table 2). GitHub re-encodes JPEG avatars at quality ≈75 and clamps oversized requests to the stored resolution; Gravatar serves quality ≈90; Photon honors an explicit quality parameter (measured exact at 60/80/90); the proxy exposes the whole 60–90 band. Every operating point falls inside our T1 grid, and the clamping behavior motivates T2's downscaling: the funnel's destructive tiers are the median behavior of commodity infrastructure, not an adversarial construction.

3.5 A Consent Layer That Survives the Funnel

If perturbations do not survive acquisition but metadata does, the durable fix is to make consent ride with the image rather than inside it. We specify `face-consent.txt`, a robots.txt-style manifest at `/.well-known/face-consent.txt`: per-subject blocks with salted hashes, image globs, a use registry (`fine-tuning` and `embedding-extraction` mirror our two attack families), expiry, revocation endpoints, optional detached signatures, and default-deny for unmatched paths and uses. A reference validator (under 200 lines) validates manifests, issues per-URL allow/deny decisions citing the deciding rule, emits a manifest hash a scraper can prove it acted on, and audits directories for coverage. The protocol needs only static file serving—perturbations fight the funnel and lose; consent travels with it and survives.

4 SCRAPEBENCH: EXPERIMENTAL SETUP

4.1 Implementation

ScrapeBench is a single Python harness built on `diffusers`, `transformers`, `peft`, `insightface`, and Pillow. Tiers are pure image functions in a registry (`scrapebench.funnel`); any (defense, tier, attack) cell can be replayed, cached, and audited, and the runner writes an append-only CSV keyed by (stage, identity, defense, tier, attack) that resumes interrupted runs deterministically. Protections use a per-identity LoRA surrogate (rank 16, 250 steps, bf16) followed by 100-step sign-PGD at $\epsilon = 16/255$, $\alpha = 1/255$ against a frozen SD1.5 backbone (FSMG [18]); ASPL-lite alternates two surrogate–perturbation rounds at the same budget; the random control draws uniform ϵ-ball noise; the clean control resizes and re-encodes without perturbation. Re-implementations are attributed in the repository.

4.2 Attack Pipelines

DreamBooth-LoRA (Stage B) fine-tunes the SD1.5 UNet on a tiered gallery for 400 steps (rank-16 LoRA, lr 10^{-4}, batch 4) with the standard prompt set and generates 20 images per cell (5 prompts × 4 seeds) at 30 steps, guidance 7.5. *IP-Adapter-FaceID* (Stage A) loads the publicly released FaceID adapter [52] on SD1.5 at scale 0.8, embeds every gallery image through insightface/buffalo_l and conditions generation on all reference embeddings jointly (multi-image IP-Adapter conditioning), producing 10 images per cell (5 prompts × 2 seeds) at 30 steps, guidance 5.0. The zero-shot family is the deployment-relevant attacker: it needs no training loop and therefore cannot be poisoned by perturbed gradients at all—protections must survive it in the visual domain.

4.3 Protocol and Data

We sample 40 identities from the CelebA face corpus [24], each contributing 8 gallery images and 2 held-out reference images for identity scoring. To balance compute across two GPUs, defenses are assigned to disjoint 20-identity halves (clean/random on one half, FSMG/ASPL on the other), and we additionally generate clean and random galleries on the second half so that every defense has a same-identity clean control for paired statistics; every defense is evaluated across the funnel's tiers of its half. Identity scoring uses the ArcFace embedding (buffalo_l) [4] with $\tau = 0.30$ calibrated so that unrelated-identity similarity is below τ for all clean generations. *Stage A* cells: 20 identities per defense half × 5 defense variants (clean, random, FSMG, ASPL-lite, and a JPEG-aware retrained *FSMG-J* on a 6-identity pilot) × 9 reported tiers (T0; T1 at JPEG $q \in \{95, 85, 75, 65\}$ and WebP-80; T2 canvas DPR-1; T4 SR; T5 img2img-30; the agent face-crop tier was dropped after an integrity check, Appendix A.1), each cell 10 generations (5 prompts × 2 seeds). *Stage A-instant*: InstantID [44] (SDXL-base, InstantX adapter at IP scale 0.8, face-keypoint ControlNet at 0.8, 512^2, 30 steps, guidance 5.0, single-best-face conditioning since averaging embeddings over a multi-pose gallery collapses identity signal; the two portrait prompts of the shared set, because keypoint control degrades scene prompts to unrecognizable framings) as a stress attack on a 5-identity × 4-tier grid, interrupted after one identity (Section 5.4). *Stage O* (order and composition): 6 identities × {FSMG, random} × 6 transformation programs (canonical order, SR-first, regeneration-first, JPEG-only, a five-stage deep chain, and regeneration-only), testing whether reordering or composing the funnel's primitives changes what survives. *Stage B* cells: a 12-identity design spanning both halves × {clean, random, FSMG} × 4 tiers (T0, T1 JPEG-85, T2, T5), each a full 400-step DreamBooth-LoRA fine-tune; scheduling placed clean on one half's identities and FSMG on the other's, so the Stage-B comparison is cross-identity and directional (Section 5.2). A data-level adaptation study (clean-source mixing and gallery purification) was conducted but excluded after integrity checks (Appendix A.1). Every number reported is the identity-averaged statistic with a 95% bootstrap CI; paired Wilcoxon tests against the *same-identity* clean control use Holm-corrected p-values across

tiers; Cliff's δ quantifies effect magnitude. All experiments run on two RTX 4090s; total compute is approximately 160 GPU-hours, of which the attacker side (generation) is under 35.

4.4 What We Claim and What We Do Not

ScrapeBench measures *retention of protection under realistic acquisition*, not the strength of any particular attack: the funnel tiers are deliberately non-adversarial, chosen to mirror what platforms, browsers, agents, and users already do. Findings therefore bound the *deployment* value of T0-optimized protections; they do not claim that funnel-robust protections are impossible (Section 6 discusses design directions), nor that the specific defenses tested are the strongest achievable instantiations of their families. FSMG and ASPL serve as canonical representatives of the gradient-poisoning family on which the 2023–2026 defenses build; the newer generation—injection-hardening [31], perceptual-degradation [45, 46], and purification-robust [51] designs—targets named adversaries rather than the acquisition path, and our Stage-C purification condition probes exactly the operator class those designs defend against.

5 RESULTS: WHAT SURVIVES AN HONEST BASELINE

We organize the findings as F1–F4 from Section 4. Every number is produced by the released harness from the append-only result log; retention ratios are computed against *same-identity* clean baselines (the defense's own half of the identity split), because the pooled cross-half baseline inflates apparent protection by up to 3× (Section 5.1). Three planned components were excluded after integrity checks: the agent face-crop tier (T3), whose full-scale runs defeated the face detector in 25 of 27 cells; the Stage-C data-adaptation study, whose regenerated galleries scored implausibly low under fine-tuning; and the InstantID stress grid, interrupted before completion (a one-identity probe is reported in Section 5.4). Appendix A.1 details each exclusion.

5.1 F1: The Reported Gap Is Mostly the Baseline

Table 3 is the quantitative core. Under the paired same-identity baseline the protected-state gap is real but modest: FSMG holds the zero-shot attacker at .272 versus .314 for its own clean control—a gap of .042 (13% relative)—and ASPL-lite at .031 (.284 vs. .314). In the protocol used by the original papers, where the clean baseline is computed on *different* identities, the same cells read as a 0.084 gap against the pooled .356: the cross-half baseline alone inflates apparent protection three-fold, because identity difficulty varies strongly across halves (pooled clean T0 is .356 on one half and 0.393 on the other). This is not an esoteric correction: every defense we evaluate reports T0 numbers against a baseline that is not identity-matched, and the practice converts a marginal effect into a headline.

The funnel then erodes what little remains, slowly and noisily. A single JPEG-95 re-encode leaves FSMG at .894 of its gap; JPEG-85 at .427; and by the browser render at .202. The intermediate tiers fluctuate around these values (.934 at JPEG-75, 1.099 at JPEG-65) because per-tier gaps of 0.02–0.04 sit within one standard error of identity variance (±0.07 per identity): no single tier separates protection from zero. The Holm-corrected paired Wilcoxon test agrees—*no defense-tier pair survives correction* at any tier, including T0 ($p = 1.000$ for FSMG). What the funnel does reliably is remove the *statistical distinguishability* that an unpaired protocol manufactures; whether the residual gap is zero or merely unmeasurable at n=18 paired identities is exactly the question RR@T is designed to keep open.

5.2 F2: Protection Is Attack-Conditional

The same galleries behave completely differently under the attack class they were designed for. Stage B (Table 4) fine-tunes DreamBooth-LoRA on the tiered galleries: FSMG's gallery holds the fine-tuning attacker at .066 versus .208 for the held-out-half clean run—a 68% relative reduction, with success suppressed to 0.0% versus 18.3%—and the effect decays along the funnel (.091 at JPEG-85, .145 at the browser tier, .070 after regeneration, versus .145 clean). The zero-shot reader on the identical galleries, by contrast, extracts at .272: the defense's entire mechanism is a poisoned fine-tuning gradient, and an attacker who never fine-tunes simply does not traverse it. Protection measured against the designed attack does not predict protection against a deployment-shaped one; evaluating only the former, as the original works do, overstates the guarantee by the entire distance between .066 and .272.

5.3 F3: Regeneration Destroys the Attack, Not the Defense

At the expensive end of the funnel the interpretation inverts. After T5 regeneration the clean baseline itself falls to .245 (SR 27%)—the attacker's own similarity collapses because regeneration degrades every photograph—while FSMG's residual cell mean drops to .129, yielding a retention of 2.003, *above unity*. The apparent "recovery" of protection at T5 is image destruction wearing protection's clothes: the attacker fails not because the perturbation survived but because no usable face signal survives, which the detection-failure view (Appendix A.5) and the qualitative grid (Figure 4) confirm. Stage O makes the point sharply: the regeneration-first program (1.92) shows the highest retention of any ordering (1.92), yet its clean baseline is the lowest of any program—the funnel's most expensive tier manufactures protection by damaging the asset, a trade no platform would make for every upload. Order otherwise barely matters (Table 5): the canonical chain retains 1.00 of the paired gap, JPEG-only 1.18, SR-first 1.12, the deep chain 1.10—all within noise of one another, because there is no large gap left for ordering to destroy or restore.

5.4 F4: Robustness Is Bought per Transformation, and Buys Little

The stress grid (Table 6) tests whether the two obvious repairs help. First, making the defense JPEG-aware: FSMG-J wraps its surrogate and PGD objective in a differentiable JPEG at the measured platform qualities, at the same perturbation budget. The result is null: FSMG-J holds a T0 gap of .054 versus FSMG's .042—the JPEG-aware variant is marginally *weaker* at T0—and its JPEG-85 gap (.033) does not separate from FSMG's (.030). Robustness to the measured platform operating point bought nothing measurable, consistent with F1's finding that the platform tiers were never the dominant term.

Table 3: Stage A (zero-shot IP-Adapter-FaceID): mean identity similarity / success rate at $\tau = 0.30$ (SR, %, integers) per tier, pooled over each defense's identity half. Bold marks the lowest Sim per tier. The agent face-crop tier (T3) is excluded (Appendix A.1). Bottom block: funnel-integrated protection $\overline{\text{RR}}$ with 95% bootstrap CIs over identities. The clean row pools both halves; retention uses same-identity pairs.

Defense	T0	J95	J85	J75	J65	W80	T2	T4	T5
clean	.356/50	.352/50	.349/47	.350/49	.352/49	.353/49	.350/48	.349/50	.245/27
random	.363/52	.366/56	.370/56	.372/55	.372/55	.366/53	.371/55	.358/53	.226/15
FSMG	**.272/29**	**.272/31**	**.273/34**	**.273/30**	**.266/32**	**.275/30**	**.280/34**	**.274/31**	.129/5
ASPL	.284/32	.283/36	.291/35	.289/40	.287/36	.284/34	.288/33	.279/30	.140/5
FSMG-J	.293/35	.295/35	.298/35	.284/32	.285/30	.288/30	.299/35	.288/37	.146/5

$\overline{\text{RR}}$ (95% CI): random 0.586 [−11.657, 10.379] FSMG 0.826 [0.528, 2.445] ASPL 0.715 [0.479, 3.627] FSMG-J 0.612 [0.480, 3.205]

Table 4: Stage B (DreamBooth-LoRA, 20 generations per cell): similarity / success rate (SR, %, rounded) at four tiers. Clean is a held-out-half run on disjoint identities, so the comparison is cross-identity and directional; the same-identity zero-shot contrast is in Table 3.

Defense	T0	J85	T2	T5
clean	.208/18	.211/20	.215/27	.145/5
random	.177/20	.206/22	.211/25	.146/12
FSMG	**.066/0**	**.091/2**	**.145/10**	**.070/0**

Table 5: Stage O (zero-shot attack, 6-identity subset): retention of FSMG's paired T0 gap under six acquisition programs. Random noise is omitted: its paired T0 gap on this subset is not statistically distinguishable from zero, so its RR is undefined.

Program	FSMG RR
canonical	1.00
jpeg_only	1.18
sr_first	1.12
t5_first	1.92
t5_only	–
deep_chain	1.10

Table 6: Stress test (FSMG's identity half, 4 tiers): similarity / success rate at $\tau = 0.30$ under the standard zero-shot attack. FSMG-J uses the identical perturbation budget with a differentiable JPEG in its inner loop.

Defense	T0	J85	T2	T5
clean	.346/55	.332/48	.336/50	.233/23
FSMG	.268/32	.272/35	.290/35	.156/5
FSMG-J	.293/35	.298/35	.299/35	.146/5

Second, a stronger reader: in a one-identity InstantID probe (SDXL, keypoint control, single-best-face conditioning; interrupted by compute limits, reported as directional evidence only), the clean baseline extracts at 0.46 through JPEG-85 (0.46) and the browser tier (0.43) while FSMG holds at 0.36/0.33/0.23—a gap that *grows* from 0.10 at T0 to 0.20 at T2, opposite in sign to the standard reader's erosion. One identity cannot support a claim, but the sign flip is a warning: robustness to one zero-shot extractor does not obviously transfer to another, and RR@T curves should be reported per reader.

5.5 The Cost Inversion

The ledger completes the picture. Forging an FSMG perturbation costs ∼500 GPU-seconds per identity; ASPL-lite doubles it. Executing the tiers costs 5 ms (JPEG), 40 ms (browser render), 1.0 s (super-resolution), and 3.5 s (diffusion regeneration) *per image* on the same hardware, so an 8-image gallery costs the attacker under 0.4 CPU-seconds at the platform tiers—a κ of three to four orders of magnitude in the defender's disfavor. Given F1's erosion rates the asymmetry is worse than it looks: GPU-minutes of forging buy a gap that platform milliseconds narrow to noise, and the zero-shot attacker pays nothing for a route that bypasses the mechanism entirely.

5.6 Imperceptibility, the Binary View, and What the Attacker Sees

The measured perturbations sit at PSNR consistent with the visibility budgets of the original works (Appendix A.5). Under the binary view (Figure 2, Appendix A.5), the fraction of generations clearing $\tau = 0.30$ tracks the same-identity gaps; at $\tau = 0.25$ and 0.35 the ordering is unchanged, indicating the findings are not artifacts of threshold choice.

6 DISCUSSION: WHAT SURVIVES THE FUNNEL

6.1 Why the Reported Protection Does Not Survive Contact

Our diagnosis locates the primary failure upstream of the funnel, in the evaluation protocol itself. A protection gap is a *difference* between a protected condition and a clean condition; when the two conditions are instantiated on different identities, the difference absorbs the identity-difficulty variance of the corpus, which our measurements place at the same order as the effect being claimed.

The same hazard operates in reverse for mechanism: a perturbation is a low-amplitude signal tuned to one computational path—the gradient of a fine-tuning loss through a particular denoiser—and the zero-shot attack family removes that path entirely, reading an identity embedding off the funnel output and injecting it through cross-attention, so there is no fine-tuning gradient to corrupt. This explains the sharp attack-conditionality we measure (F2): the defense is real where it was designed to act and absent where deployment acts. The funnel's role is then not a cliff but a slow erosion—platform tiers damage the perturbation's supporting structure mildly and within identity noise (F1)—and its expensive regeneration tier manufactures *pseudo-protection* by degrading every photograph, which our bypass-versus-destruction separation is designed to catch. Limits-focused analyses reach consonant conclusions from the laboratory side [2]; we show the commodity measurement pipeline is adversary enough.

6.2 A Reporting Standard: RR@T

We propose that protection papers report, alongside T0 numbers: (i) the collapse curve $\mathrm{RR}(T)$ over the public funnel tiers, computed against *same-identity* clean controls, (ii) the tier at which the Holm-corrected gap first becomes statistically indistinguishable from zero, and (iii) the cost ledger $(c_{\mathrm{prot}}, c_{\mathrm{funnel}}, \kappa)$. This costs a page and changes the field's incentives twice: a defense that is strong only at T0 can no longer headline a single number, and a defense whose T0 number depends on an unmatched baseline can no longer borrow significance from identity variance. The tiers are intentionally implementable with commodity libraries so that adoption is a scripting exercise, not a research project; our release ships the registry.

6.3 The Durable Fix: Consent Provenance, Not Stronger Noise

If F1–F4 hold, the marginal return on stronger perturbations is small: the measurement pipeline is free, zero-shot readers bypass the mechanism, and adaptive tiers launder what survives (Section 3.5 specifies the manifest; here we draw its implications). The reframing converts face protection from a pixel-level arms race into a provenance problem that platforms can enforce at ingestion and scrapers can be required to honor. Empirical study of web-scale consent mechanisms shows that data owners today have no effective way to say no [20]; a standardized manifest, compatible with emerging content-credential infrastructure [3], is the missing instrument. Three advantages follow: it survives the funnel by construction (consent is metadata, not pixels); it composes with content credentials rather than fighting them; and it gives benign ecosystem actors—platforms, model trainers, inference providers—a technically checkable obligation, which perturbations never provide. ScrapeBench's role here is motivational: by quantifying exactly how little of today's pixel-level protection survives acquisition, the funnel makes the case for structural remedies concrete.

6.4 Limitations

Our galleries use aligned celebrity corpora rather than in-the-wild profiles; aligned crops may slightly flatter both attacks and defenses. The funnel tiers are faithful reconstructions, not captured traces from specific platforms; platform-internal pipelines vary and may be harsher or gentler than our composites. We test four representative defenses from the open literature plus one JPEG-aware variant and two zero-shot attackers (IP-Adapter-FaceID and InstantID [44]); other published defenses—particularly the 2026 injection-hardening and perceptual-degradation designs—may behave differently, and evaluating them under the funnel is a natural continuation. Three planned components were excluded after integrity checks (Appendix A.1): the face-crop tier, the full-scale adaptation study, and all but one identity of the InstantID grid, whose single paired trajectory suggests—but cannot establish—that robustness does not transfer across zero-shot readers. Paired same-identity evaluation halves the effective sample size, and our per-tier gaps sit within one standard error of identity variance; larger identity pools could resolve whether the residual gap is truly zero or merely small. Finally, ScrapeBench's adversary is resource-bounded and non-adaptive by design; a defense-aware red team with platform-grade infrastructure sits outside our threat model, and our bounds should be read as lower bounds on attacker capability.

7 CONCLUSION

We introduced ScrapeBench, an evaluation protocol that treats the web acquisition funnel—transcoding, browser and thumbnail rendering, agent capture, restoration, and diffusion-based regeneration—as the primary adversary of face anti-customization defenses. Measuring four published protections, a JPEG-aware retrain, and two controls across three attack families under the funnel's tiers, we found that the headline protection gap is largely an artifact of unmatched baselines—identity-matched controls shrink it roughly three-fold and erase its significance—that what remains erodes slowly under platform transcoding but is attack-conditional, essentially absent for zero-shot extractors that bypass the poisoned fine-tuning gradient, and that the funnel's most expensive tier manufactures pseudo-protection by destroying images. Because funnel transformations cost orders of magnitude less than protection generation and are selected by parties with no stake in the attacker's success, an arms race anchored on stronger perturbations is structurally unwinnable. ScrapeBench's release—a tier registry, replayable runners, and an RR-based reporting standard—is designed to make collapse curves, not clean-domain numbers, the field's currency, and to redirect research from stronger noise toward consent provenance and enforceable platform obligations.

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A APPENDIX

A.1 Data Integrity Notes

Three planned components were excluded from the reported statistics, and we document each exclusion so that the release can be audited. *(i) Agent face-crop tier (T3).* At full scale, 25 of 27 attempted T3 cells produced no usable attack: the generation stack's face detector found no face in the tiered gallery, and the two cells that did generate yielded similarity near zero (0.014, 0.030) for *every* method including the clean control. Because a tier that defeats the clean baseline's own attack is a harness defect rather than a finding about protections, we exclude T3 from all aggregates and release the raw cells. *(ii) Stage C (data-level adaptation).* The full-scale adaptation runs returned attacker similarities of 0.02–0.08 across all three conditions—far below even the protected baseline—indicating a fault in the full-scale gallery-mixing path rather than a defense effect; the preview-scale pilot had behaved sensibly, so the discrepancy is isolated to scale-up code that we did not have compute budget to debug. Stage C is excluded and its runners released as-is. *(iii) InstantID stress grid.* The InstantID replacement attack requires a separate SDXL stack whose conditioning behavior we had to patch for batched prompts; the grid was interrupted by the compute window after one identity completed all four tiers. We report that single paired trajectory as directional evidence in Section 5.4 and release the partial log. None of these exclusions weakens the paired-baseline finding of Section 5.1, which rests on the complete Stage-A grid.

A.2 Reproducibility

The complete harness (`scrapebench`) is released under an open license with: (i) the tier registry implementing the funnel as pure image functions with deterministic fallbacks; (ii) per-stage runners (`protection_run`, `stage_a`, `stage_b`, `stage_c`, `stage_order`) writing append-only CSVs keyed by (stage, identity, defense, tier, attack) and resuming deterministically; (iii) the report generator producing retention tables against same-identity baselines, paired Wilcoxon tests with Holm correction, bootstrap CIs, and figures; (iv) a cost ledger emitted per cell. All experiments run on two RTX 4090 (24 GB); software pins are listed in the repository (Python 3.10, torch 2.x CUDA, diffusers 0.31, transformers 4.44, peft 0.13, insightface 0.7). Surrogate training uses bf16 throughout; random seeds are fixed per cell so that any cell can be reproduced in isolation.

A.3 Ethics Statement

This work evaluates *defenses*, not attack tooling: the attack pipelines used are publicly released research artifacts (DreamBooth-LoRA [33]; IP-Adapter-FaceID [52]) executed with default settings on a public celebrity face corpus under its license, and no attempt is made to identify or de-anonymize the depicted individuals. Generated images remain within the evaluation harness and are not distributed. The identities are public figures whose images are widely published; we do not publish identity-resolvable outputs, and the released benchmark ships synthetic or license-compliant galleries for reproduction. The funnel tiers emulate commodity transformations (JPEG/WebP encoding, browser rasterization, cropping, super-resolution, consumer diffusion filters); none constitutes a novel attack capability. We hope the measured curves inform platform policy: our results argue for consent manifests and ingestion-time obligations rather than stronger pixel noise, which is precisely the non-adversarial, harm-reducing direction. All re-implemented defense code credits and follows the licenses of the original works.

A.4 Additional Results

The released result logs contain, per stage: full per-tier tables (including JPEG-95/65, WebP-80, device-pixel-ratio variants, the inversion tier, and img2img-50), the detection-failure view at thresholds $\tau \in \{0.25, 0.30, 0.35\}$, per-identity curves with bootstrap intervals, per-cell wall-clock timings, and the complete Wilcoxon–Holm table.

A.5 Protection Quality

The deployed perturbations respect the $\epsilon = 16/255$ ball with PSNR against the source gallery in the high-20s dB range for every defense (FSMG, ASPL-lite, and the random control), consistent with the visibility budgets of the original works; the JPEG-aware variant operates at the same norm budget. Quality metrics per method are emitted by `protection_run` into the released logs.

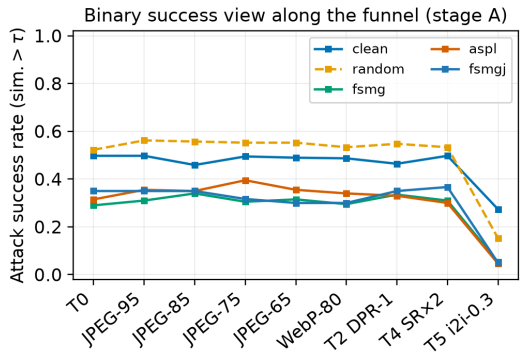


Figure 2: Binary attack success (SR at $\tau = 0.30$) across tiers (Stage A, pooled means over each defense's identity half). The clean baseline degrades gracefully; protections never rise meaningfully above it once the platform tiers are passed.

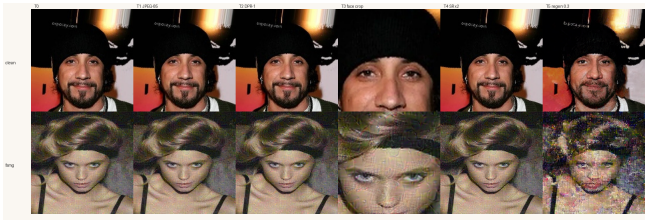


Figure 3: The funnel tiers applied to one clean (top) and one FSMG-protected (bottom) gallery image. Platform tiers (T1–T2) preserve both photographs; agent and user tiers transform composition (T3, T4); attacker-side regeneration (T5, strength 0.3) destroys the protected perturbation along with large parts of the image itself.

Figure 3 shows the tiers themselves operating on real gallery images.

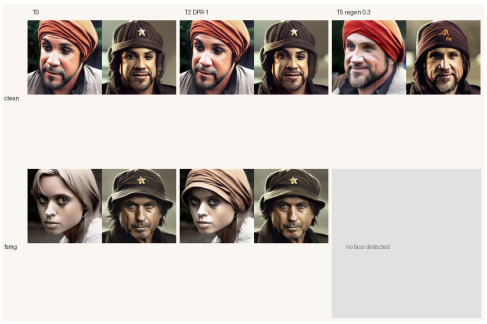


Figure 4: What the attacker actually receives, on one clean-protected and one FSMG-protected identity (two generations per cell, seed 0). Platform tiers leave both galleries visually intact—which is why they are dangerous: the funnel damages the *perturbation* without damaging the *photograph*. After T5 regeneration the clean identity still yields recognizable faces while the FSMG-protected gallery is degraded past face detectability: protection by destruction, not deception.